

Information loss of fixed linear health indicators as a control problem: simplex geometry and reachability limits

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Abstract

A health indicator compresses several condition-monitoring channels into one scalar, and normally keeps the same weights for the life of the component. We take it to be a fixed linear map of noise-whitened channels on which every mechanism acts positively, over a population whose units differ only in where they sit on one degradation path, with remaining life Gaussian across it. Because degradation runs through several mechanisms at differing rates, the channel combination carrying most information about remaining life rotates as the component ages, so a fixed indicator necessarily discards information. We show that the operating point sets how much it discards, which makes that loss a control variable. The loss reduces to a single angle, between the indicator and the direction along which information currently arrives, and writing each channel's share of that information as a probability vector places that angle on a statistical simplex. Zero loss is attainable exactly when the directions reachable at each instant share a common point, which for generic plants needs one fewer independently adjustable operating variable than there are mechanisms. Scaling every channel by the same factor leaves the loss unchanged, so no criterion built from the Fisher information about remaining life can detect it. Steering the operating point reduces the loss substantially when the authority it offers is large beside the drift of the informative direction; on a representative three-mechanism synthetic plant

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the information-loss objective defined here falls by a factor of 68 against the best constant operating point. A dimensionless index of that balance, available before any indicator is chosen, describes the regime rather than deciding it, with an area under the receiver-operating curve of 0.73 over 299 random plants. Six public run-to-failure fleets test the premises separately: an informative direction estimated from measured cells rotates, and simulated turbofan records confirm against a negative control that the operating point moves it. The same fleets bound the theory: the highest-fidelity turbofan data reject the separable plant form under which the operator’s authority becomes computable offline, so that branch describes a model class rather than measured machinery.

Keywords: prognostics, health indicator, remaining useful life, information loss, optimal control, reachability

1. Introduction

Condition monitoring rarely feeds raw measurements to a life-prediction algorithm. A vibration channel, a temperature, an impedance and a capacity reading first go into a single scalar, the health indicator, and the life-prediction algorithm sees only that scalar [1, 2]. The compression is almost always fixed: the designer picks one set of weights at design time, and every unit carries those weights for its whole life, because firmware embeds them, certification freezes them, or re-tuning them in the field is impractical.

Fixing the compression has a cost. Real components degrade through several mechanisms at once, and those mechanisms do not proceed in step: an early-life process may dominate for the first third of the record and then saturate, while a late-life process accelerates. The combination of channels that carries the most information about remaining life therefore rotates as the component ages. A fixed indicator is aligned with that combination at best at one moment, and is misaligned for the rest of the record; the information it discards is not recoverable downstream, however good the prediction algorithm that follows [3].

The established response is to make the map better: choose a different criterion for it, let it switch between segments of life, or learn it from more run-to-failure data [4, 2]. All of these take the degradation as given and work on the compression. This paper acts on the degradation itself, through the operating point.

23 The rotation is not a property of the instrumentation. The degradation
24 physics produces it, and that physics responds to stress: temperature, load
25 and duty cycle change the relative rates of the competing mechanisms, not
26 only the overall speed. Running a cell hotter does not only age it faster;
27 it also shifts the balance between the mechanisms, and therefore moves the
28 direction in which information about remaining life arrives.

29 The operating point is, in most applications, something the operator
30 chooses. So the misalignment between a fixed indicator and the informa-
31 tive direction is not fixed by nature. It is a function of the input, and the
32 operator can steer it.

33 That reframing turns a data-analysis question into a control question, and
34 control questions come with sharper answers than design heuristics do. One
35 can ask what misalignments are reachable at a given operating envelope,
36 whether zero loss is attainable at all, how many independent inputs that
37 would take, and what the best achievable loss is when it is not. These are
38 the questions this paper answers.

39 They sit against five bodies of work. Four address parts of the problem
40 and the fifth supplies the geometry used in Section 3; to our knowledge none
41 of them addresses the position taken here.

42 The prognostics literature treats the compression as a design problem to
43 be solved from data: searching genetically over candidate features against
44 monotonicity and trendability criteria [2], learning it with a recurrent net-
45 work [5], or surveying the resulting pipelines [1, 6]. Two limits on that
46 programme are known, information-theoretic bounds on what any fixed com-
47 pression can retain [3] and the number of run-to-failure units needed to learn
48 one reliably [4]. The first is our own, and the boundary is worth drawing
49 explicitly: it takes one degradation trajectory as given and asks how much
50 information a fixed compression must lose along it. The question here is the
51 complementary one, whether that trajectory can itself be steered so that the
52 same fixed map loses less. Both prior limits hold the trajectory fixed; neither
53 asks whether it can be changed.

54 The predictors downstream are overwhelmingly stochastic — Wiener,
55 gamma and inverse-Gaussian degradation processes with first-passage fail-
56 ure [7, 8], reviewed for remaining-life estimation in [9, 10] and embedded in
57 system-level design methodology in [11]. That literature models how damage
58 evolves, not how the measurement should be compressed before the model is
59 applied, which is the question upstream of it.

60 A separate literature does treat the operating point as a decision variable,

Table 1: Where this paper sits. The columns are the axes that separate these literatures, not a ranking: each body of work answers a question the others do not ask, and the last column is a restriction this paper imposes on itself rather than an advantage over the others. “Input alters degradation” means the operating point enters the degradation dynamics; “direction” means the informative direction of Section 2 is itself an objective.

	input alters degradation	lifetime is an objective	direction is an objective	indicator fixed, scalar
Indicator design [2, 5, 1]	—	—	—	✓
Fixed-indicator limits [3, 4]	—	—	—	✓
Stochastic life models [7, 9]	—	—	—	—
Health-aware control [12, 13]	✓	✓	—	—
Observability-aware [14, 15, 17]	✓	—	partly	—
This paper	✓	secondary	✓	✓

but to extend remaining life rather than to preserve estimability [12, 13]; the two aims can conflict, and Section 6 exhibits an instance in which they disagree. Choosing inputs to make a state easier to estimate is itself classical, going back to dual control [14], and its modern forms shape a scalarisation of the Fisher information matrix [15, 16, 17]. Our Proposition 4.6 shows the quantity at issue here to be invisible to every such scalarisation.

The fifth body is information geometry. The reformulation of Section 3 rests on the Fisher–Rao geometry of the probability simplex and on the Bhattacharyya coefficient [18, 19], and the flow it produces is the replicator equation of evolutionary dynamics [20, 21]. These are standard objects; what is new here is their identification with the informative direction of a degrading component, and the fact that the fitness driving the flow is actuated.

The contributions are as follows.

1. The information a fixed indicator discards is an exact function of one misalignment angle (Lemma 2.2). Placing the channels’ shares of the Fisher information on a statistical simplex turns the degradation into a replicator flow whose rates the operating point modulates, and the optimal indicator into a barycentre of the degradation path (Section 3).
2. Zero loss is attainable exactly when the reachable direction sets share a point, which generically needs at least as many independent operating variables as mechanisms less one (Section 4). Supplying the missing input collapses the peak reachability floor from 4.52° to zero at every state, so the rule prices instrumentation rather than only bounding it. Taking logarithms makes every reachable set a translate of one fixed set,

which turns the question into a containment and shows the counting rule to be sufficient but not necessary; the exceptions exist for any number of mechanisms and are too fragile to build on (Section 4.1). The count separates two objectives that are easily conflated: holding the misalignment at a constant level is a single scalar condition and generically needs one input, whereas freezing the informative direction outright is $d - 1$ conditions. That gap is the difference between useful steering and lossless steering, and an installation with one controllable variable can have the first without the second.

3. The loss is invariant to a common gain on all channels and is therefore invisible to every scalarisation of the Fisher information about remaining life, so no criterion built from that information alone can rank indicators by the loss they incur.
4. The best achievable loss is the value of an optimal control problem whose objective is a ratio, and whose terminal time is itself controlled (Section 5); we compute its value to two digits and do not certify it. A dimensionless regime index, computable before any indicator is designed, describes the regime in which steering can have substantial value, and its accuracy and the direction of its errors are measured on 299 randomly drawn plants (Section 6.7).
5. The premises are separately testable on public run-to-failure data, and Section 7 tests three in turn: whether the informative direction rotates, whether the operating point moves it, and whether the separable stress model survives. The first holds on measured fleets, the second so far only in simulation, and the third is rejected. That section therefore reports which hypotheses withstand contact with data rather than demonstrating that the method works.

The results do not all rest on the same assumptions, and it is worth separating them before any of them is stated. Lemma 2.2, Proposition 2.3 and Theorem 4.1 need only the observation model of Section 2: a fixed linear indicator on whitened channels, and a scalar latent. Lemma 4.3 and Proposition 4.4 need more, namely that the input enters each mechanism through a single separable factor, and Section 7 reports that the highest-fidelity turbfan data reject exactly that. The empirical readings of Section 7 need more again, since estimating one informative direction from a fleet presumes the units differ by a time shift rather than by their degradation coefficients. Fig. 6 is the map, and the reader who wants to know which branch a given

122 result sits on should consult it rather than infer it.

123 Section 2 states the model. Sections 3 to 5 develop the theory, Section 6
 124 exercises the three of its claims that are stated sharply enough to be refuted,
 125 and Section 7 reports what public fleet data can and cannot establish about
 126 the premises the theory rests on.

127 2. Problem formulation

128 This section states the degradation model, says precisely what a prognos-
 129 tic algorithm is being asked to estimate and where the uncertainty in that
 130 quantity comes from, derives the direction along which information about
 131 it arrives, and defines the fixed indicator whose information loss the rest of
 132 the paper studies. Readers familiar with [3] will recognise the model; the
 133 derivation is given in full here because the geometry of Section 3 depends on
 134 several of its details.

135 2.1. Degradation driven by a controlled operating point

136 A component degrades through d distinct mechanisms acting at once. In
 137 a lithium-ion cell these might be loss of cyclable lithium, growth of the solid-
 138 electrolyte interphase and mechanical fatigue of the electrode; in a rolling
 139 bearing, spalling, wear and lubricant degradation. We write $x_i(t) \geq 0$ for the
 140 damage accumulated by mechanism i up to time t , and collect these into the
 141 state $x = (x_1, \dots, x_d)$.

142 One externally chosen operating point drives all d mechanisms. It collects
 143 the temperature, the mechanical or electrical load, and the duty cycle at
 144 which the operator runs the component. We write that operating point as

$$u(t) \in \mathcal{U} \subset \mathbb{R}^m, \quad (1)$$

145 with $\mathcal{U}$ compact, representing the operating envelope the application permits.
 146 Throughout this paper u is the control input: it is the decision variable,
 147 and the question we ask is what choosing it well can buy. The number of
 148 independent inputs, m , turns out to decide what is achievable, and it is
 149 usually much smaller than the number of mechanisms d .

150 The damage states evolve according to

$$\dot{x}_i = \varphi_i(u) h_i(x_i), \quad i = 1, \dots, d, \quad (2)$$

151 in which the two factors play different roles:

- 152 • $\varphi_i(u) > 0$ is the stress-acceleration factor of mechanism i : how much
153 faster that mechanism runs at operating point u than at a reference
154 point. It carries all the dependence on the input.
- 155 • $h_i(x_i) > 0$ is the intrinsic shape of mechanism i : how its rate depends
156 on the damage it has already done. It carries all the dependence on
157 the state.

158 This separation is the standard accelerated-life form and covers the stress
159 models in routine use [22]. In the Arrhenius model the input is absolute
160 temperature and $\varphi_i(u) = A_i \exp(-E_i/(ku))$, with $A_i > 0$ a pre-exponential
161 constant, $E_i > 0$ the activation energy of mechanism i and k Boltzmann's
162 constant. In the inverse-power-law model the input is a mechanical or electri-
163 cal stress and $\varphi_i(u) = \kappa_i u^{n_i}$, with $\kappa_i > 0$ a scale factor and $n_i > 0$ the stress
164 exponent. Choosing h_i recovers the exponential, power-law and stretched-
165 exponential degradation classes [3]; the results below use only $\varphi_i > 0$ and
166 $h_i > 0$.

167 The within-unit dynamics are deterministic, where most of the degrada-
168 tion literature places the randomness inside a single unit's path [7, 8, 10].
169 We put it between units instead (Section 2.2). The object of study is the
170 direction along which information arrives, which is fixed by the relative rates
171 of the competing mechanisms: a within-unit diffusion would jitter that di-
172 rection without altering its systematic motion, whereas dispersion in the rate
173 parameters across a population is what makes remaining life uncertain and
174 therefore worth estimating.

175 We measure damage on a scale normalised so that mechanism i exhausts
176 the component's tolerance when x_i reaches unity. A component is withdrawn
177 at a condemnation limit at or below that point, so we write the end of life
178 as the first passage of any one mechanism to a threshold $x_f \in (0, 1]$:

$$T_f = \inf\{t \geq 0 : \max_i x_i(t) \geq x_f\}. \quad (3)$$

179 No result below depends on the value of x_f , which enters only by fixing where
180 a trajectory stops. The computed quantities of Section 6 do depend on it,
181 and the value used there is stated with the rest of the test plant.

182 2.2. What is to be estimated, and why it is uncertain

183 The quantity a prognostic system must report is the remaining useful life

$$R(t) := T_f - t, \quad (4)$$

184 the time left before failure.

185 We must say where the uncertainty in R lies, because (2) is a deterministic
 186 differential equation: for one component whose initial damage and parame-
 187 ters were known exactly, T_f and hence R would be known exactly, and there
 188 would be nothing to estimate. The uncertainty is between components, not
 189 within one. Units built to the same specification differ in initial damage and
 190 in the coefficients of (2), which is the unit-to-unit variability that reliability
 191 engineering treats as the primary source of scatter in life data [22]. A prog-
 192 nostic algorithm observes one unit drawn from that population and must
 193 infer its remaining life.

194 We therefore treat R , at any given observation instant, as a random
 195 variable over the population of units, with prior variance $\text{Var}[R] > 0$. This
 196 variance is the uncertainty the algorithm exists to reduce, and it is the reason
 197 the estimation problem is not trivial. Everything below is written for a
 198 nominal trajectory of (2), with the fleet spread entering only through $\text{Var}[R]$.

199 2.3. Observations and the informative direction

200 Each damage state reaches the record through its own noisy channel,

$$y_i = x_i + \sigma_i \epsilon_i, \quad i = 1, \dots, d, \quad (5)$$

201 with $\sigma_i > 0$ the noise level of channel i and ϵ_i independent standard normal
 202 variables. The record available to the algorithm is the collection of these
 203 measurements over the observation window.

204 Equation (5) pairs each mechanism with a channel of its own. Condition
 205 monitoring in general returns $y = H(x, u) + \epsilon$, with sensors that mix mech-
 206 anisms, and the pairing is an assumption rather than a description. It is
 207 load-bearing: the positivity of $\mathbf{d}$ that Remark 3.1 shows the results to need
 208 follows from $\varphi_i > 0$ and $h_i > 0$ in these coordinates and need not survive
 209 an arbitrary H . Where H is known, invertible and positivity-preserving the
 210 construction carries over to $H^{-1}y$; the general case does not reduce to it.

211 Two assumptions carry the sensitivity, and both are stated here rather
 212 than left to be inferred.

213 *The prior on R .* We take R to be Gaussian over the population with variance
 214 $\text{Var}[R]$. Linearising the record about the nominal trajectory, following [3],
 215 then makes the record and R jointly Gaussian, and the mutual informations
 216 below are exact. Linearisation alone would not do it: with a non-Gaussian
 217 prior of the same variance the Gaussian value is an upper bound on what

218 a channel carries, and Lemma 2.2 becomes an inequality rather than an
 219 identity. The information geometry that follows is exact under this Gaus-
 220 sian prior and the time-shift population described next; it is not a model of
 221 population variability in general.

222 *How units differ.* We take the population spread to be a reparametrisation
 223 of a single degradation path: units differ in where they sit on the trajectory
 224 of (2), not in its shape. This is what makes R a coordinate along that path,
 225 and it is what the next display uses. Dispersion in the coefficients of (2) is a
 226 different matter, because it moves $\partial x/\partial \theta$ rather than sliding along $\partial x/\partial R$,
 227 and Remark 2.1 records what it costs.

228 The sensitivity of channel i to remaining life then follows from (4): be-
 229 cause R falls at unit rate as t advances,

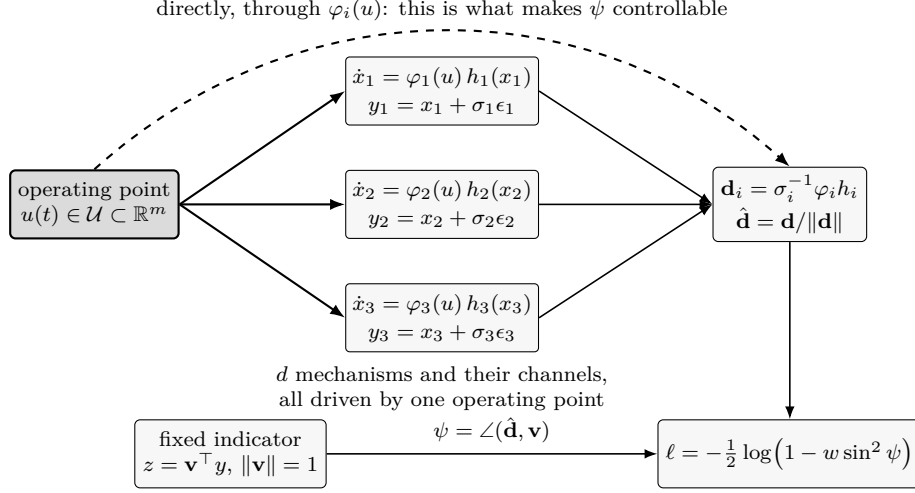
$$\frac{\partial x_i}{\partial R} = -\frac{\partial x_i}{\partial t} = -\varphi_i(u) h_i(x_i). \quad (6)$$

230 **Remark 2.1** (what the time-shift assumption costs). *Equation (6) is exact*
 231 *when the population spread slides along one path, and only then. On the*
 232 *plant of Section 6.1, with units instead drawn by dispersing the degradation*
 233 *coefficients, the direction that the population regression of x on R actually*
 234 *returns stands about 45° from $-\varphi(u)h(x)$, across two families of dispersed*
 235 *coefficient and every spread tried; a control whose units are started on the*
 236 *reference trajectory, where (6) holds by construction, returns the same re-*
 237 *gression to within 0.2° . The misalignments this paper is about are a few*
 238 *degrees. Extending the construction to populations dispersed in θ , where the*
 239 *observation carries scatter that no value of R explains, is open.*

240 Dividing each channel by its own noise level puts them on a common
 241 scale, and taking magnitudes gives the whitened signal direction

$$\mathbf{d}(t) = \left[\sigma_i^{-1} \varphi_i(u) h_i(x_i) \right]_{i=1}^d, \quad \hat{\mathbf{d}} = \mathbf{d}/\|\mathbf{d}\|. \quad (7)$$

242 In words: $\mathbf{d}$ says how strongly, and in what combination of channels, the
 243 measurement responds to a change in remaining life. Its direction $\hat{\mathbf{d}}$ is the
 244 *informative direction*, the combination a health indicator has to be aligned
 245 with and the object named informally in Section 1. Its length $\|\mathbf{d}\|$ is the
 246 strength of that signal. The two objects carry the paper: $\|\mathbf{d}\|$ is how much
 247 information arrives and $\hat{\mathbf{d}}$ is where it arrives from, and the loss of a fixed



Alignment asks $\hat{\mathbf{d}}(x, u) = q$: $d - 1$ scalar equations on m unknowns, whence the counting rule of Theorem 4.1.

Figure 1: The construction. The operating point enters twice, through the plant and directly through $\varphi_i(u)$; the second path is what makes the misalignment ψ a control variable. One input drives all d mechanisms.

indicator depends only on the second. Experiment design in the observability tradition scores the first, which is why Proposition 4.6 finds it blind to the loss.

We use two properties of (7) repeatedly below.

Positivity. Every entry of $\mathbf{d}$ is strictly positive, because $\varphi_i > 0$ and $h_i > 0$ in (2). This is a consequence of the degradation model rather than an assumption imposed for convenience, and Remark 3.1 shows that the results below would fail without it.

Actuation. $\mathbf{d}$ is an explicit algebraic function of the state and the input. The direction along which information about R arrives is therefore not a fixed property of the instrumentation: it moves when the operating point moves. That is the observation this paper is built on, and Fig. 1 places it: the operating point reaches $\hat{\mathbf{d}}$ twice, once through the state and once directly.

Because R is scalar, a single signal-to-noise ratio governs the information the record carries about it, namely the squared sensitivity times the prior uncertainty,

$$\gamma = \|\mathbf{d}\|^2 \text{Var}[R]. \quad (8)$$

264 Rates in (7) are taken with respect to clock time, whereas [3] differentiates
 265 with respect to age normalised by T_f . The two conventions differ by a factor
 266 common to all d channels, so they give the same $\hat{\mathbf{d}}$, and hence the same
 267 geometry in Sections 3–4, while rescaling γ . Section 5, where lifetime enters
 268 the objective, restores the normalised convention.

269 2.4. The fixed health indicator and its information loss

270 A health indicator compresses the d measured channels into one scalar
 271 before any life prediction is made. We take it to be a fixed linear map.
 272 Since (7) already measures each channel in units of its own noise, we write
 273 the indicator in those same coordinates: with $\Sigma := \text{diag}(\sigma_1^2, \dots, \sigma_d^2)$ and
 274 $\tilde{y} := \Sigma^{-1/2}y$ the whitened record,

$$z = \mathbf{v}^\top \tilde{y}, \quad \|\mathbf{v}\| = 1, \quad (9)$$

275 with the weight vector $\mathbf{v}$ chosen once, at design time, and held for the whole
 276 life of the component. Fixing it is not a modelling convenience: the map
 277 is typically embedded in firmware or frozen by certification, so it cannot be
 278 re-tuned as the component ages. An indicator specified on the raw measure-
 279 ments, $z = \mathbf{w}^\top y$, is the same map under $\mathbf{v} \propto \Sigma^{1/2}\mathbf{w}$, and the normalisation
 280 $\|\mathbf{v}\| = 1$ only fixes the scale of z ; it restricts nothing. The distinction matters
 281 because the angle below is measured after whitening: unless the σ_i are equal,
 282 a unit vector in raw coordinates is not a unit vector in whitened ones, and
 283 the two readings of ψ disagree.

284 We compute the cost of fixing $\mathbf{v}$ as follows. Write $\psi(t)$ for the angle
 285 between the informative direction $\hat{\mathbf{d}}(t)$ and the indicator weights $\mathbf{v}$. Af-
 286 ter whitening, the signal in the record points along $\hat{\mathbf{d}}$ and has magnitude
 287 $\|\mathbf{d}\|\sqrt{\text{Var}[R]}$, while the noise is isotropic of unit scale. Projecting onto a unit
 288 vector $\mathbf{v}$ therefore keeps the fraction $\cos \psi$ of the signal and all of the noise,
 289 so the indicator’s signal-to-noise ratio is

$$\gamma \cos^2 \psi \quad (10)$$

290 against the γ of (8) available to an oracle that is free to re-aim its projection at
 291 $\hat{\mathbf{d}}(t)$ at every instant. Since a scalar Gaussian channel of signal-to-noise ratio
 292 s carries $\frac{1}{2} \log(1 + s)$ nats about R [23], the information the fixed indicator
 293 forfeits, at each instant, is

$$\ell = \frac{1}{2} \left[\log(1 + \gamma) - \log(1 + \gamma \cos^2 \psi) \right] \geq 0, \quad (11)$$

294 The design objective throughout is its average over the life of the component,

$$\bar{\ell} := \frac{1}{T_f} \int_0^{T_f} \ell(t) dt, \quad (12)$$

295 a loss per unit time. We optimise this rate rather than the information the
 296 whole observation record forfeits, because (12) is unaffected by how densely
 297 the record is sampled and is therefore a property of the indicator rather
 298 than of the measurement schedule. The two are different quantities, and
 299 Proposition 2.3 gives the second in closed form.

300 A third quantity is the one an operator actually pays for: the error of
 301 whatever remaining-life estimator runs downstream. It is neither of the first
 302 two and is not a function of them. The loss is a logarithmic information
 303 quantity and the error is a time, and an estimator holding a whole history
 304 recovers much of what any single instant discards, so a large spread in (12)
 305 does not imply a comparable spread in error. What the theory can offer
 306 downstream is therefore an ordering rather than a conversion, and Section 6.8
 307 tests that ordering against an estimator built from none of this construction:
 308 it survives, with a spread in error much smaller than the spread in loss.
 309 Equation (12) should be read accordingly, as an information-geometric ob-
 310 jective whose ranking carries downstream, not as a surrogate for prognostic
 311 accuracy.

312 Two remarks on what (11) does and does not say. It is a loss relative to
 313 an oracle that no fixed map can realise, so it measures the price of the time-
 314 invariance constraint rather than of a bad design; a perfectly chosen $\mathbf{v}$ still
 315 pays it whenever $\hat{\mathbf{d}}$ moves. And it depends on the indicator only through the
 316 single angle ψ : two indicators that make the same angle with $\hat{\mathbf{d}}$ are equally
 317 good, whatever else distinguishes them.

318 The next lemma replaces (11) by a closed form that will be more conve-
 319 nient throughout, and records how it behaves at small misalignment.

320 **Lemma 2.2.** *Let $w := \gamma/(1 + \gamma) \in (0, 1)$. Then*

$$\ell = -\frac{1}{2} \log(1 - w \sin^2 \psi), \quad (13)$$

321 *exactly and at every signal-to-noise ratio, and*

$$\ell = \frac{w}{2} \psi^2 + \frac{w(3w-2)}{12} \psi^4 + O(\psi^6). \quad (14)$$

322 *Proof.* Substituting $\cos^2 \psi = 1 - \sin^2 \psi$ into (11) gives

$$\ell = -\frac{1}{2} \log \frac{1 + \gamma - \gamma \sin^2 \psi}{1 + \gamma} = -\frac{1}{2} \log (1 - w \sin^2 \psi).$$

323 For (14), use $-\frac{1}{2} \log(1-s) = \frac{s}{2} + \frac{s^2}{4} + O(s^3)$ with $s = w \sin^2 \psi$ and $\sin^2 \psi = \psi^2 -$
 324 $\psi^4/3 + O(\psi^6)$; collecting the ψ^4 terms gives $-w/6 + w^2/4 = w(3w-2)/12$. $\square$

325 The quadratic term of (14) is the bound obtained in [3, Thm. 1], which is
 326 therefore the leading term of an exact identity rather than an approximation
 327 of unknown quality. The quartic coefficient $w(3w-2)/12$ changes sign at
 328 $w = 2/3$, that is at $\gamma = 2$: below that signal-to-noise ratio the quadratic
 329 model overstates the loss, above it the model understates it. This matters in
 330 practice because prognostic problems sit on both sides of $\gamma = 2$: an early-life
 331 measurement with a weak signal on one side, a well-instrumented component
 332 near failure on the other. Fig. 2(a) plots the exact loss against its quadratic
 333 approximation on both sides of that ratio.

334 **Proposition 2.3.** *Let the record consist of measurements (5) at instants*
 335 *$t_1, \dots, t_N$, with the noise independent across channels and across instants,*
 336 *and write $\mathbf{d}_k := \mathbf{d}(t_k)$ and ψ_k for the angle between $\hat{\mathbf{d}}_k$ and $\mathbf{v}$. Put*

$$G := \text{Var}[R] \sum_{k=1}^N \|\mathbf{d}_k\|^2, \quad W := \frac{G}{1+G},$$

$$S := \frac{\sum_k \|\mathbf{d}_k\|^2 \sin^2 \psi_k}{\sum_k \|\mathbf{d}_k\|^2}.$$

337 *Then the information about R that the fixed indicator forfeits over the whole*
 338 *record is*

$$\ell_{\text{rec}} = -\frac{1}{2} \log (1 - WS). \quad (15)$$

339 *Proof.* See Supplementary Section S2. $\square$

340 Equation (15) is (13) again, with the instantaneous signal-to-noise ratio
 341 γ replaced by its sum G over the record, and $\sin^2 \psi$ by the mean S weighted
 342 by the signal energy $\|\mathbf{d}_k\|^2$. The reduction to one angle therefore survives the
 343 passage from an instant to a record. What does not survive is the averaging:
 344 $\bar{\ell}$ averages the loss, whereas (15) averages $\sin^2 \psi$ inside it, and the two differ.
 345 They also select different indicators. Minimising (15) over $\|\mathbf{v}\| = 1$ maximises
 346 $\mathbf{v}^\top \left(\sum_k \mathbf{d}_k \mathbf{d}_k^\top \right) \mathbf{v}$, so the best fixed indicator for the record is the leading

eigenvector of the integrated Fisher matrix, while the minimiser of $\bar{\ell}$ is the barycentre of Proposition 3.3. Every result below is stated for $\bar{\ell}$.

Corollary 2.4 (what the loss costs an estimator). *In the model of Section 2, the minimum-variance estimate of R formed from the fixed indicator carries mean-squared error*

$$\frac{\text{Var}[R \mid z]}{\text{Var}[R \mid \tilde{y}]} = \frac{1 + \gamma}{1 + \gamma \cos^2 \psi} = e^{2\ell}$$

times that of the estimate formed from the whole whitened record, at each instant. The same identity holds over a record with ℓ replaced by ℓ_{rec} of Proposition 2.3.

Proof. Write $\tilde{y} = \mathbf{d}R + \epsilon$ with $\epsilon \sim N(0, I_d)$ and R Gaussian. The indicator returns $z = (\mathbf{v}^\top \mathbf{d})R + \mathbf{v}^\top \epsilon$, whose noise is standard normal because $\|\mathbf{v}\| = 1$, so its signal-to-noise ratio is $\gamma \cos^2 \psi$ and $\text{Var}[R \mid z] = \text{Var}[R]/(1 + \gamma \cos^2 \psi)$. The record's sufficient statistic is $\mathbf{d}^\top \tilde{y}$, with ratio γ , giving $\text{Var}[R \mid \tilde{y}] = \text{Var}[R]/(1 + \gamma)$. Their quotient is $e^{2\ell}$ by (11). The record statement is the same computation with G and S in place of γ and $\sin^2 \psi$. $\square$

The loss is therefore not a score standing in for estimation error: it is the logarithm of the factor by which that error inflates. What the corollary does not settle is whether the ordering survives an estimator other than the one the model prescribes, and Section 6.8 checks that separately.

3. The information simplex and its replicator flow

Lemma 2.2 reduced the loss to one angle. This section shows that the angle lives on a familiar object, the probability simplex, and that the degradation traces a flow along it that the operating point drives. The payoff is threefold: the loss becomes a recognised statistical distance, the optimal indicator becomes a barycentre, and the effect of the input becomes a single interpretable quantity. Two of the results here are load-bearing for what follows — the replicator form of Theorem 3.4 and the misalignment rate of Theorem 3.6, which Section 4 builds the reachability argument on. The Bhat-tacharyya identity and the barycentre are interpretation: they say what the loss and the optimal indicator already are in recognised terms, and nothing later depends on them.

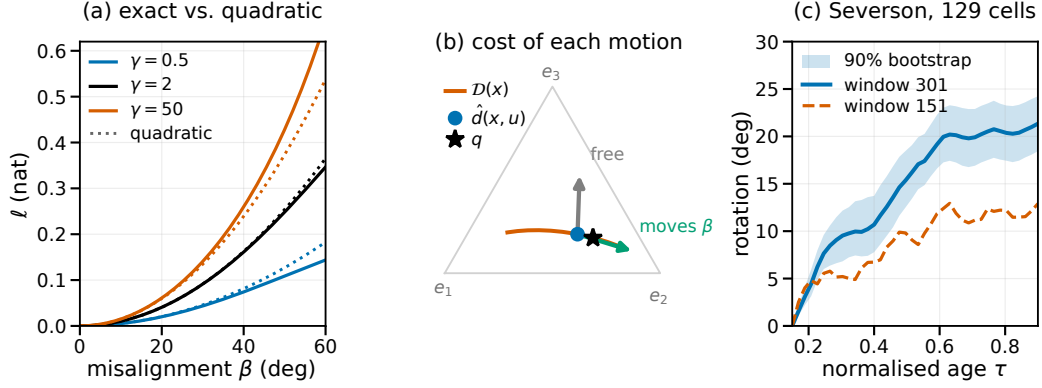


Figure 2: (a) Lemma 2.2: the exact loss (solid) against its quadratic model (dotted), whose quartic coefficient changes sign at $\gamma = 2$. (b) Theorem 3.6: of the replicator velocity, only the component along the geodesic towards q moves β . (c) The rotation, measured on the Severson fleet at two differentiation windows.

3.1. Information shares

Because every entry of $\mathbf{d}$ is positive (Section 2.3), the squared components of the unit vector $\hat{\mathbf{d}}$ form a probability vector:

$$p_i := \hat{\mathbf{d}}_i^2, \quad p_i > 0, \quad \sum_i p_i = 1. \quad (16)$$

The interpretation is direct. The total Fisher information the record carries about R is $\|\mathbf{d}\|^2 \text{Var}[R]$ by (8), and channel i contributes $\mathbf{d}_i^2 \text{Var}[R]$ of it. So p_i is exactly the share of the total information about remaining life that arrives through channel i . A component in which one mechanism dominates the degradation has p near a vertex of the simplex; one in which three mechanisms contribute equally sits near its centre.

The same map applied to the indicator, $q_i := \mathbf{v}_i^2$, is legitimate because an optimal indicator may be taken with non-negative weights.

Remark 3.1 (positivity is load-bearing). *If $\hat{\mathbf{d}} \geq 0$ then $\hat{\mathbf{d}}^\top |\mathbf{v}| \geq |\hat{\mathbf{d}}^\top \mathbf{v}|$ at every instant, so replacing $\mathbf{v}$ by $|\mathbf{v}|$ never increases (11): non-negative weights are without loss of generality. This argument uses the positivity of $\mathbf{d}$ inherited from $\varphi_i, h_i > 0$, and it fails without it: in random trials with sign-indefinite $\hat{\mathbf{d}}$, taking absolute values was worse than the signed vector about a third of the time. The results therefore require the positive-orthant structure of the degradation model; it is not only a notational convenience.*

395 3.2. The loss as a Bhattacharyya distance

396 The identification below is used twice in what follows: it gives “the aver-
 397 age of a curve on the simplex” an unambiguous meaning (Section 3.3), and
 398 it lets the high-signal limit of the loss be read off as a known divergence.

399 **Proposition 3.2.** $\cos \psi = \sum_i \sqrt{p_i q_i}$ is the Bhattacharyya coefficient of p and
 400 q . Consequently $\psi = \frac{1}{2} d_{\text{FR}}(p, q)$, where d_{FR} is the Fisher–Rao distance on
 401 the simplex, and, writing $D_{\text{B}}(p, q) = -\log \sum_i \sqrt{p_i q_i}$ for the Bhattacharyya
 402 divergence,

$$\ell = -\frac{1}{2} \log \left[(1 - w) + w e^{-2D_{\text{B}}(p, q)} \right]. \quad (17)$$

403 As $\gamma \rightarrow \infty$, $\ell \rightarrow D_{\text{B}}(p, q)$.

404 *Proof.* By construction,

$$\cos \psi = \hat{\mathbf{d}}^\top \mathbf{v} = \sum_i \sqrt{p_i} \sqrt{q_i},$$

405 which is precisely the Bhattacharyya coefficient of p and q . The map $p \mapsto \sqrt{p}$
 406 is the classical isometry between the simplex under the Fisher–Rao metric
 407 and the positive orthant of the unit sphere, under which the geodesic distance
 408 is twice the spherical angle [18, 19]; hence $\psi = \frac{1}{2} d_{\text{FR}}$. Then $\cos^2 \psi = e^{-2D_{\text{B}}}$,
 409 and substituting into (13) gives (17). Letting $w \rightarrow 1$ gives the limit. $\square$

410 The square root is a bijection between the simplex and the non-negative
 411 part of the unit sphere, carrying p to $\hat{\mathbf{d}}$ and q to $\mathbf{v}$, and by the proposition
 412 it is an isometry. We use the two descriptions interchangeably from here
 413 on: a statement such as $q \in \mathcal{D}(x)$ in Section 4 says that $\mathbf{v}$ is a reachable
 414 direction, and a distance between points of the simplex is the angle between
 415 the corresponding unit vectors.

416 The resulting picture is the one the rest of the paper develops. The
 417 degrading component is a curve on the simplex, the health indicator is a
 418 single point on it, and the information lost depends only on the distance
 419 between them. Design of the indicator is the choice of one point; control of
 420 the operating point is the shaping of the curve.

421 3.3. The optimal indicator as a barycentre

422 If the indicator is one point and the plant is a curve, the best point should
 423 be some kind of average of the curve. It is, but of two different kinds in two
 424 different limits.

425 **Proposition 3.3.** *Let q^* minimise the time-averaged loss.*

- 426 (i) *As $\gamma \rightarrow \infty$, $\ell \rightarrow D_B$ pointwise, so q^* tends to the Bhattacharyya*
 427 *barycentre $\arg \min_q \int D_B(p_\tau, q) d\tau$.*
 428 (ii) *As the curve shortens, $\ell = \frac{w}{2}\psi^2 + O(\psi^4)$, so q^* tends to the w -weighted*
 429 *Fisher–Rao barycentre, and $\bar{\ell} = \frac{1}{8}\bar{w} \text{Var}^{\text{FR}} + O(\psi^4)$ where $\text{Var}^{\text{FR}} :=$*
 430 *$\min_q \mathbb{E}_\mu[d_{\text{FR}}^2(p_\tau, q)]$ is the μ -weighted Fréchet variance of the curve.*

431 *Proof.* See Supplementary Section S2. □

432 The two limits are taken in different parameters, signal-to-noise ratio in
 433 (i) and arc length in (ii), and they do not coincide. On a curve of angular
 434 span 82.7° the two barycentres lie 0.72° apart, and q^* crosses from the Fisher–
 435 Rao side to the Bhattacharyya side between 0 and 10 dB, reaching 3×10^{-5}
 436 degrees of the latter by 50 dB. For design purposes the practical consequence
 437 is mild: either barycentre of a nominal degradation path is a good starting
 438 indicator.

439 3.4. The replicator flow and the role of the input

440 We now ask how fast the informative direction moves and what the op-
 441 erating point does to that motion.

442 **Theorem 3.4.** *Let $g_i = \log \mathbf{d}_i$ denote the logarithmic rate of channel i .*
 443 *Along (2),*

$$\dot{\hat{\mathbf{d}}}_i = \hat{\mathbf{d}}_i(\dot{g}_i - \mathbb{E}_p[\dot{g}]), \quad (18)$$

444 *so the information shares obey the replicator equation of evolutionary dynam-*
 445 *ics, in which each share grows in proportion to the amount by which its own*
 446 *rate exceeds the average, the average being taken over the shares themselves:*

$$\dot{p}_i = 2p_i(\dot{g}_i - \mathbb{E}_p[\dot{g}]), \quad (19)$$

447 *and*

$$\|\dot{\hat{\mathbf{d}}}\| = \text{Std}_p[\dot{g}], \quad \dot{g} = \left[\varphi_i(u) h'_i(x_i) \right]_i + \nabla_u \log \varphi \cdot \dot{u}. \quad (20)$$

448 *Proof.* From $\mathbf{d}_i = e^{g_i}$ we have $\dot{\mathbf{d}}_i = \mathbf{d}_i \dot{g}_i$, so $\dot{\hat{\mathbf{d}}} = \|\mathbf{d}\|^{-1}(\dot{\mathbf{d}} - \hat{\mathbf{d}}(\hat{\mathbf{d}}^\top \dot{\mathbf{d}}))$ has
 449 components $\hat{\mathbf{d}}_i \dot{g}_i - \hat{\mathbf{d}}_i \sum_j \hat{\mathbf{d}}_j^2 \dot{g}_j$, which is (18); squaring and summing gives
 450 $\text{Var}_p[\dot{g}]$. Since $p_i = \hat{\mathbf{d}}_i^2$, $\dot{p}_i = 2\hat{\mathbf{d}}_i \dot{\hat{\mathbf{d}}}_i = 2p_i(\dot{g}_i - \mathbb{E}_p[\dot{g}])$. Differentiating $g_i =$
 451 $\log \varphi_i(u) + \log h_i(x_i) - \log \sigma_i$ along (2) gives $\dot{g}_i = \nabla_u \log \varphi_i \cdot \dot{u} + (h'_i/h_i)\varphi_i h_i$,
 452 and $(h'_i/h_i)\varphi_i h_i = \varphi_i h'_i$. □

Equation (20) has a compact reading. The speed at which the informative direction rotates equals the information-weighted spread of the mechanisms' logarithmic rates. If all mechanisms accelerate at the same proportional rate the direction does not move at all, however fast the component is degrading; it is disagreement among the mechanisms that rotates it. The identity is the Fisher–Rao/Shahshahani geometry of the simplex combined with Fisher's fundamental theorem of natural selection [18, 20, 21]. What is new here is its identification with the informative direction of a degrading component, and, decisively, that the quantity playing the role of fitness, the logarithmic rate $\dot{g}_i$ of channel i , is actuated: the second summand in (20) is the operator's lever.

Remark 3.5 (contrast with Oja's flow). *Oja's flow for online principal-component extraction has the same algebraic form; for a diagonal covariance it reduces to a replicator equation with fitness twice the eigenvalues [24]. The roles are opposite. There the flow is an algorithm, run to fixation in order to find a direction, with constant fitness. Here it is the plant, its fitness is time-varying and actuated, and the aim is to prevent fixation from carrying the direction away from a fixed indicator.*

Only one scalar matters for the loss. Fix the indicator q and write $\beta := \angle(\hat{\mathbf{d}}, \mathbf{v}) = \frac{1}{2}d_{\text{FR}}(p, q)$ for the misalignment. This is the angle ψ of Lemma 2.2 under a new name, and the change of name marks a change of role: ψ was a fixed angle between two directions, whereas $\beta = \beta(x_t, u_t)$ is followed along the trajectory and is the quantity the input acts on. Every loss expression below may therefore be read as a function of β .

Theorem 3.6. *Let the operating-point trajectory $u(\cdot)$ be absolutely continuous, and let $\hat{e}$ be the unit tangent at $\hat{\mathbf{d}}$ along the geodesic towards $\mathbf{v}$. Then $\dot{\beta} = -\langle \dot{\hat{\mathbf{d}}}, \hat{e} \rangle$, which is affine in the input rate: $\dot{\beta} = \alpha_0(x, u) + \alpha(x, u)^\top \dot{u}$. In particular $|\dot{\beta}| \leq \|\dot{\hat{\mathbf{d}}}\|$, with equality only for motion along that geodesic.*

Proof. $\cos \beta = \hat{\mathbf{d}}^\top \mathbf{v}$ gives $-\sin \beta \dot{\beta} = \dot{\hat{\mathbf{d}}}^\top \mathbf{v}$. Also $\hat{e} = (\mathbf{v} - \cos \beta \hat{\mathbf{d}}) / \sin \beta$ and $\dot{\hat{\mathbf{d}}}^\top \hat{\mathbf{d}} = 0$, so $\langle \dot{\hat{\mathbf{d}}}, \hat{e} \rangle = \dot{\hat{\mathbf{d}}}^\top \mathbf{v} / \sin \beta = -\dot{\beta}$. Affinity follows because $\dot{g}$ is affine in $\dot{u}$ by (20) and $\hat{\mathbf{d}}$ is linear in $\dot{g}$. $\square$

The hypothesis is needed only to write $\dot{u}$; nothing in Section 4 depends on it, since the reachable sets (21) are defined pointwise in u and the limits

below hold for every measurable admissible input. Theorem 3.6 is a local kinematic description of smooth steering, not a restriction on the controls the rest of the paper admits.

The component of the replicator velocity orthogonal to $\hat{e}$ is therefore free: the informative direction may rotate in a $(d - 2)$ -dimensional subspace at no cost in information. Fig. 2(b) shows the decomposition at one state. This is the observation that separates the two objectives of the next section.

4. Reachability limits on the misalignment

Controlling the rate $\dot{\beta}$ is not the same as reaching a value of β . The distinction is the subject of this section, and it is what decides whether a fixed indicator can be made lossless at all.

For a state x , the set of informative directions the operator can produce by choosing the operating point is

$$\mathcal{D}(x) := \{\hat{\mathbf{d}}(x, u) : u \in \mathcal{U}\} \subset S^{d-1}, \quad (21)$$

the reachable direction set, and the closest the indicator can be brought at that state is the reachable misalignment floor

$$\beta_{\min}(x, q) := \text{dist}(q, \mathcal{D}(x)). \quad (22)$$

For a single input, $\mathcal{D}(x)$ is an arc traced out as u sweeps $\mathcal{U}$; its length measures the actuation authority available at x , and it shrinks to a point when the mechanisms respond identically to the input.

Theorem 4.1. *Fix an admissible input and let $x(\cdot)$ be the trajectory it generates. Zero loss along that trajectory is attainable with the fixed indicator q if and only if $q \in \bigcap_t \mathcal{D}(x_t)$. If $m < d - 1$ and $u \mapsto \hat{\mathbf{d}}(x, u)$ is continuously differentiable, the indicators that attain zero loss form a null subset of the unit sphere, so almost every q , and in fact every q that does not attain it, has $\beta_{\min} > 0$ on a set of times of positive measure. Whether any q at all attains zero loss is a property of the plant, and Remark 4.5 exhibits plants for which one does. In every case $\bar{\ell} \geq \int \ell(\beta_{\min}(x_t, q)) dt / T_f$ for that trajectory.*

Proof. By Lemma 2.2, $\ell(\beta) = 0$ iff $\sin \beta = 0$, so $\bar{\ell} = 0$ iff $\beta_t = 0$ for almost every t , that is iff $\hat{\mathbf{d}}(x_t, u_t) = q$, that is iff $q \in \mathcal{D}(x_t)$ for every t . Solving $\hat{\mathbf{d}}(x, u) = q$ imposes $d - 1$ independent scalar equations, both sides being

unit vectors, on m unknowns. If $m < d - 1$ then $\mathcal{D}(x)$, the image of an m -dimensional set under a continuously differentiable map, is a null subset of the $(d - 1)$ -dimensional sphere, which almost every q misses; if $m \geq d - 1$ the implicit function theorem supplies a solution wherever the Jacobian has full rank. The inequality $\beta_t \geq \beta_{\min}(x_t, q)$ holds for every admissible input by definition of $\beta_{\min}$ as a minimum over $\mathcal{U}$, and is strict exactly where $q \notin \mathcal{D}(x_t)$. Since $\mathcal{U}$ is compact and $\hat{\mathbf{d}}$ continuous, each $\mathcal{D}(x_t)$ is compact and varies continuously with t , so the set of times at which $q \notin \mathcal{D}(x_t)$ is open. It is non-empty for every q that does not attain zero loss, in particular for every q outside the null set $\mathcal{D}(x_0)$, and so has positive measure. Since ℓ is increasing on $[0, \pi/2]$, the bound follows on dividing by T_f . $\square$

This is a state-conditioned result. The bound is genuine, in the sense that no input violates a reachability constraint at the state it occupies, but it is conditional on the trajectory, which the input also steers. Optimising over the trajectory as well is deferred to Section 5, and the two levels are the structure of the argument: a state-conditioned limit here, a trajectory-level limit there.

The practical reading of Theorem 4.1 is a counting rule: a component degrading through d mechanisms generically needs $d - 1$ independently adjustable operating variables before a fixed indicator can be kept perfectly aligned. Here *generically* is a count of dimensions over the plant's data, not a statement of measure: the theorem proves the measure statement over indicators, Remark 4.5 constructs plants that escape the count, and we do not prove those plants rare. Section 4.1 says what the qualifier covers. For a generic plant under this model, three mechanisms and a single temperature setpoint will not do it. The three results of this section answer different questions, and it is worth separating them. Theorem 4.1 characterises zero loss exactly, as membership of every reachable set, and then counts dimensions to say when a generic plant can meet it; the count is necessary for generic data and is not by itself a construction. Proposition 4.4 gives an exact criterion, necessary and sufficient, on the separable subclass alone. Remark 4.5 exhibits plants outside that subclass which attain zero loss with fewer inputs than the count asks, so the count is not necessary in general. But a weaker and much cheaper objective is available.

Corollary 4.2 (what actuation is actually required). *Two objectives that are easily conflated carry different requirements.*

- 551 (i) Driving $\bar{\ell}$ to zero is equivalent to freezing $\hat{\mathbf{d}}$ at q , and needs $d - 1$
 552 conditions on $\dot{u}$, hence generically $m \geq d - 1$.
 553 (ii) Holding β at a constant level, not necessarily zero, is the single scalar
 554 equation $\alpha_0 + \alpha^\top \dot{u} = 0$, solvable whenever $\alpha \neq 0$ and the resulting $u(\cdot)$
 555 stays in $\mathcal{U}$; generically $m \geq 1$ suffices.

556 *Proof.* (i) By Theorem 4.1, $\bar{\ell} = 0$ iff $\hat{\mathbf{d}}(x_t, u_t) = q$ for every t , so $\hat{\mathbf{d}}$ is constant;
 557 by Theorem 3.4 that forces $\dot{g}_i = \mathbb{E}_p[\dot{g}]$ for every i , which is $d - 1$ independent
 558 conditions on $\dot{u} \in \mathbb{R}^m$. (ii) is one equation in $\dot{u}$ by Theorem 3.6. The gap
 559 between them is the content of that theorem: $\dot{\beta}$ sees only the component of
 560 $\dot{\hat{\mathbf{d}}}$ along $\hat{e}$, so the remaining $d - 2$ directions of rotation are unconstrained by
 561 (ii) and constrained by (i). $\square$

562 Minimising $\bar{\ell}$ is neither of these: it trades the level of β against the
 563 trajectory that carries it, and is the problem of Section 5.

564 4.1. The reachable sets are translates of one set

565 Theorem 4.1 counts equations against unknowns and hedges with
 566 “generic”. The hedge is not idle, and the following observation says exactly
 567 what it covers.

568 **Lemma 4.3** (separability). *Write $\xi(x, u)$ for the class of $\log \mathbf{d}(x, u)$ modulo*
 569 *the all-ones vector, which is what $\hat{\mathbf{d}}$ retains. Then*

$$\begin{aligned} \xi(x, u) &= a(x) + b(u), \\ a_i(x) &= \log \frac{h_i(x_i)}{\sigma_i}, \quad b_i(u) = \log \varphi_i(u), \end{aligned} \tag{23}$$

570 *both taken modulo the all-ones vector. Consequently every reachable direction*
 571 *set is the same set translated: $\mathcal{D}(x) = a(x) + \mathcal{B}$ with $\mathcal{B} := b(\mathcal{U})$ fixed.*

572 *Proof.* Immediate from $\mathbf{d}_i = \sigma_i^{-1} \varphi_i(u) h_i(x_i)$ in (7): the logarithm turns the
 573 product into a sum, and the normalisation of $\hat{\mathbf{d}}$ removes any component along
 574 the all-ones vector. $\square$

575 The decomposition separates the two parties. The set $\mathcal{B}$ is fixed by the
 576 stress model φ and the envelope $\mathcal{U}$ alone (Fig. 3(a)). An operator’s authority
 577 can therefore be computed offline from the accelerated-life model, with no
 578 degradation record at all. The curve $\mathcal{A} := \{a(x_t)\}$ is fixed by the damage
 579 shapes h and the noise levels σ , and every reachable set is that one set carried

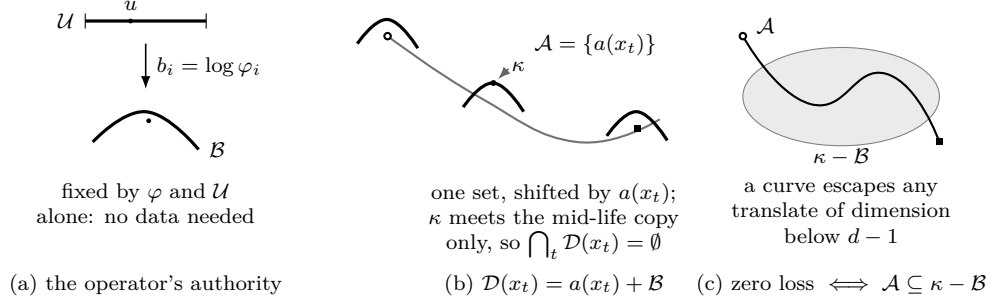


Figure 3: How the counting rule becomes a containment. (a) The operator's authority $\mathcal{B} = b(\mathcal{U})$, fixed by the stress model and the envelope alone. (b) Every reachable set is that one set translated, $\mathcal{D}(x_t) = a(x_t) + \mathcal{B}$ (circle: start of life, square: failure). (c) Zero loss asks the whole curve to lie in one translate (Proposition 4.4).

580 along it (Fig. 3(b)). What the geometry of Section 3 measures is how far the
 581 plant's demand $\mathcal{A}$ sticks out of the operator's authority $\mathcal{B}$.

582 That offline route is the most useful consequence drawn in this paper
 583 and also the most restrictive. It holds on the separable subclass and not
 584 beyond it, and the best high-fidelity turbofan data we have reject separability
 585 (Section 7). This subsection and Proposition 4.4 should therefore be read as
 586 describing a structured subclass that is convenient when it applies, not as
 587 a general account of real machinery. Fig. 6 marks the one branch of the
 588 development that depends on it.

589 **Proposition 4.4** (exact criterion). *Zero loss is attainable along a trajectory*
 590 *if and only if $\mathcal{A}$ is contained in a translate of $-\mathcal{B}$. In particular $\mathcal{A}$ must lie*
 591 *in an affine subspace of dimension at most $\dim \mathcal{B} \leq m$, which for a curve*
 592 *in the $(d-1)$ -dimensional quotient fails generically unless $m \geq d-1$; this*
 593 *recovers Theorem 4.1. When it does not fail, $m < d-1$ suffices.*

594 *Proof.* By Theorem 4.1, zero loss holds iff $q \in \mathcal{D}(x_t)$ for every t . Writing κ
 595 for the quotient class of $\log \mathbf{v}$ and using Lemma 4.3, that reads $\kappa - a(x_t) \in \mathcal{B}$
 596 for every t , that is $\mathcal{A} \subseteq \kappa - \mathcal{B}$. The dimension statement follows because a
 597 translate of $\mathcal{B}$ spans an affine subspace of dimension $\dim \mathcal{B}$. $\square$

598 The containment is drawn in Fig. 3(c), and the mechanism the count
 599 obscures is now visible. Alignment at an instant fixes u_t outright, since
 600 $b(u_t) = \kappa - a(x_t)$ determines it; the input is spent, and any residual constraint
 601 on the state must then hold along the flow with no freedom left to enforce

602 it. One input can carry the whole burden only when no residual constraint
603 is left, which happens when $\mathcal{A}$ already lies parallel to $\mathcal{B}$.

604 **Remark 4.5** (a family with $m = 1$ and any d). *Take exponential shapes*
605 *$h_i(x) = e^{\nu_i(x_i - x_i^0)}$, which (2) admits, and Arrhenius factors. Under the ramp*
606 *$u(t) = u_0 + rt$ the damage rates are constant, $x_i(t) = x_i^0 + \varphi_i^0 t$ with $\varphi_i^0 =$*
607 *$\varphi_i(u_0)$, so $a(x_t)$ moves at constant velocity $\nu_i \varphi_i^0$. Choosing $\nu_i = rE_i/\varphi_i^0$ makes*
608 *that velocity parallel to E , which is the direction of $\mathcal{B}$, and $b(u_t)$ retreats*
609 *along $\mathcal{B}$ at the same rate. The two cancel and $\hat{\mathbf{d}}$ is frozen. Supplementary*
610 *Section S4 integrates the construction for $d = 2$ to 8.*

611 This bears on the sharpness of Theorem 4.1. The family of Remark 4.5
612 is a knife edge. At one percent of error in its shape coefficients, steering
613 is already worse than not steering (Supplementary Section S4). A designer
614 should read Proposition 4.4 as an explanation of when the counting rule
615 binds, and Theorem 4.1 as the rule to design against.

616 4.2. Invisibility to Fisher-information criteria

617 A natural objection is that all of this should follow from designing the in-
618 put to maximise the Fisher information about R , as in the observability-aware
619 literature discussed in Section 1. It does not, and the reason is structural.

620 **Proposition 4.6** (Fisher blindness). *The loss of a fixed indicator is not a*
621 *function of the Fisher information about R : configurations with equal Fisher*
622 *information can have different losses, and configurations with equal losses*
623 *can have different Fisher information. No criterion depending on the Fisher*
624 *information alone can therefore rank fixed indicators by the loss they incur.*

625 *Proof.* For a scalar latent the Fisher information is the number $\|\mathbf{d}\|^2 \text{Var}[R]$,
626 so trace, determinant and least eigenvalue coincide and every scalarisation is
627 a strictly monotone function of it. Under the rescaling $\mathbf{d} \mapsto c\mathbf{d}$ with $c > 0$,
628 that number changes while $\hat{\mathbf{d}}$, and hence β and ℓ , do not. A common gain
629 therefore equalises the scores of two configurations with different losses, and
630 separates the scores of two configurations with the same loss. $\square$

631 Stated in engineering terms, a Fisher criterion measures the strength of
632 the signal, whereas the loss of a fixed indicator depends on its direction.
633 Amplifying every channel by the same factor increases the first and leaves
634 the second unchanged. Section 6 exhibits this concretely: with the Fisher
635 score held equal to seven digits, the loss still spans a factor of 9.4.

636 5. The best achievable loss as a fractional control problem

637 Theorem 4.1 bounds the loss along a given trajectory. But the input
 638 steers the trajectory as well as the misalignment on it, so the best achievable
 639 loss requires optimising over both at once. This changes the structure of the
 640 problem.

641 The objective is a time average over a lifetime that the input itself deter-
 642 mines:

$$\bar{\ell}^*(q) = \min_{u(\cdot) \in \mathcal{U}} \frac{\int_0^{T_f} \ell(\beta(x_t, u_t)) dt}{T_f}. \quad (24)$$

643 Both the integrand and the upper limit T_f depend on $u(\cdot)$. This makes
 644 (24) a fractional optimal control problem, meaning that the objective is a
 645 ratio rather than that the dynamics carry a fractional-order derivative. The
 646 fraction is not a technicality: it is the source of the trade-off, as follows.

647 Suppose T_f were fixed. Nothing in (2) penalises the input directly, so
 648 the answer would be trivial: find whichever admissible u minimises β and
 649 stay there. Because T_f is itself controlled, the denominator of (24) rules that
 650 solution out. The operating point that aligns the indicator is in general not
 651 the operating point that prolongs life, so aligning now shortens the interval
 652 over which the alignment is averaged. The optimum must trade the two,
 653 and Section 6.5 measures how steep that trade is on a concrete plant. A
 654 fixed-horizon formulation represents none of it, which is why one was tried
 655 and discarded (Section 6.5).

656 5.1. A converse bound uniform over trajectories

657 Before computing anything we record a certificate that removes the con-
 658 ditioning. Theorem 4.1 bounds the loss along a trajectory the input has
 659 already chosen; the following proposition bounds it over all of them at once,
 660 and it is what would make the number computed below a limit rather than
 661 an observation, were a function satisfying it exhibited; none is, for reasons
 662 given below.

663 **Proposition 5.1** (verification certificate). *Fix an indicator q and a constant*
 664 *$\lambda \geq 0$. Suppose there is a continuously differentiable V on a neighbourhood*
 665 *of the admissible state region such that*

666 (i) *$V(x) \leq 0$ for every x on the failure surface $\{\max_i x_i = 1\}$, and*

667 (ii) writing $D_i(x, u) = \varphi_i(u)h_i(x_i)$ for the drift of (2), for every admissible
 668 state x and every $u \in \mathcal{U}$,

$$\ell(\beta(x, u)) - \lambda + \nabla V(x)^\top D(x, u) \geq 0. \quad (25)$$

669 Then every admissible input satisfies $\bar{\ell} \geq \lambda + V(x_0)/T_f$. In particular, if
 670 $V(x_0) \geq 0$ then $\bar{\ell}^*(q) \geq \lambda$, uniformly over all admissible inputs and hence
 671 over all trajectories.

672 *Proof.* Let $u(\cdot)$ be any admissible input and x_t its trajectory. Since V is C^1
 673 and $t \mapsto x_t$ is absolutely continuous, $t \mapsto V(x_t)$ is absolutely continuous and
 674 the chain rule gives $\frac{d}{dt}V(x_t) = \nabla V(x_t)^\top \dot{x}_t = \nabla V(x_t)^\top D(x_t, u_t)$ for almost
 675 every t . Applying (25) at (x_t, u_t) gives $\ell(\beta(x_t, u_t)) - \lambda + \frac{d}{dt}V(x_t) \geq 0$ a.e.,
 676 and integrating over $[0, T_f]$,

$$\int_0^{T_f} (\ell - \lambda) dt \geq V(x_0) - V(x_{T_f}) \geq V(x_0),$$

677 the last step by (i), since x_{T_f} lies on the failure surface. Dividing by $T_f > 0$
 678 gives the claim. $\square$

679 Condition (25) constrains the plant data alone: it quantifies over every
 680 state and every admissible input, and mentions no trajectory. Any V one
 681 can exhibit therefore certifies a limit, and the search for a converse becomes
 682 the search for a verification function.

683 The C^1 hypothesis is not a formality, and it is exactly what the nat-
 684 ural candidates lack: the value function of the Dinkelbach problem be-
 685 low is only Lipschitz and violates (25) pointwise, while affine functions are
 686 smooth but certify only a vacuous pointwise bound. Supplementary Sec-
 687 tion S3 records both attempts and what they imply, that a certificate must
 688 be smooth and must have a gradient that varies, which points at a smooth
 689 under-approximation of the value function by sum-of-squares programming
 690 or interval arithmetic. We do not claim a certified converse here, and the
 691 optimum computed below remains a computation; what Proposition 5.1 buys
 692 is the replacement of an open problem without shape by a concrete object
 693 with a checkable property.

694 Problems of the form (24) are handled by Dinkelbach's transform [25]:
 695 setting

$$F(\lambda) = \min_{u(\cdot) \in \mathcal{U}} \int_0^{T_f} (\ell - \lambda) dt, \quad (26)$$

the map F is decreasing in λ and vanishes exactly at the optimal ratio, so λ^* is found by bisection on $F(\lambda) = 0$. The inner problem in (26) has free terminal time and an absorbing set, hence a stationary Hamilton–Jacobi–Bellman equation on the state alone, with no time argument. Because $\dot{x} > 0$ the state advances monotonically, and a semi-Lagrangian scheme [26] converges by value iteration. One implementation detail matters: the scheme must use a fixed spatial step rather than a fixed time step, because degradation speeds vary by orders of magnitude across $\mathcal{U}$ and a fixed time step is unusable at one end of the envelope or the other.

6. Numerical study

Sections 3–5 are analytical, and three of their claims are stated sharply enough to be tested rather than illustrated: that the loss is invisible to a Fisher criterion, that the counting rule $m \geq d - 1$ binds in both directions on a generic plant, and that the best achievable loss is a finite number one can compute. This section instantiates the model on a synthetic three-mechanism plant on which every quantity is available in closed form, checks those three claims against it, measures what steering is worth, and reads off the dimensionless index that says in advance whether it will repay the effort. Section 7 then reports what measured fleets can and cannot establish about the premises.

6.1. The test plant

We take $d = 3$ mechanisms with Arrhenius factors $\varphi_i(u) = e^{\theta_i - E_i u}$, in which the input u is the reciprocal temperature $1/(kT)$, so that this is the Arrhenius form of Section 2 with e^{θ_i} the pre-exponential constant A_i . We set $\theta = (0, 0.3, -0.2)$ and $E = (0.6, 1.1, 0.8)$, and shapes $h_i(x) = 1 + c_i x$ with $c = (-0.5, 2.0, 0.8)$, giving one decelerating and two accelerating mechanisms. There is a single input $u \in [0, 3]$, so $m = 1 < d - 1 = 2$, and Theorem 4.1 puts zero loss out of reach unless this plant is one of the exceptions of Remark 4.5; Section 6.3 computes the floor and finds that it is not. Noise is unit and the prior variance is $\text{Var}[R] = 10^3$, so the weight $w = \gamma/(1 + \gamma)$ of Lemma 2.2 is evaluated at the plant’s own $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ at each state rather than held fixed; γ varies along a trajectory because $\|\mathbf{d}\|$ does. Every trajectory starts from $x_i(0) = 5 \times 10^{-3}$ and runs to the condemnation limit $x_f = 0.9$ of (3). Raising x_f to unity leaves every qualitative conclusion below intact and moves the constants: the peak floor of Section 6.3 rises from 4.52° to 5.41° ,

Table 2: Loss of the fixed indicator at a Fisher score held equal. The exposure factor δ , defined in the text, holds $\delta^2 \|\mathbf{d}\|^2$ fixed and leaves $\hat{\mathbf{d}}(\tau)$ unaffected. Losses use $\gamma = \|\mathbf{d}\|^2 \text{Var}[R]$ with $\text{Var}[R] = 10^3$.

u	δ	$\delta^2 \ \mathbf{d}\ ^2$	rms β (deg)	$\bar{\ell}$ (nat)
0.0	0.235	0.4307565	6.23	0.0215
1.0	0.628	0.4307565	4.53	0.0123
1.5	1.000	0.4307565	6.22	0.0208
2.0	1.564	0.4307565	7.53	0.0379
3.0	3.769	0.4307565	6.57	0.1161

and the gain of Section 6.5 falls from 68 to 58. No constant reported below should be read across thresholds.

Verification of the identities. The analytical identities were checked numerically before use, and the residuals are reported in Supplementary Section S4.

6.2. A Fisher level set of varying loss

Proposition 4.6 is easy to demonstrate on this plant. For a constant input, a common gain on $\mathbf{d}$ moves $\|\mathbf{d}\|$ while leaving $\hat{\mathbf{d}}(\tau)$, and hence β , untouched: the Fisher score and the loss can be varied independently. Write $\delta(u)$ for the gain that equalises the score across inputs, fixed by $\delta(u)^2 \|\mathbf{d}\|^2(u) = F$ with F its value at $u = 1.5$. It is realised by rescaling how long the component runs under load, which is why it does not rotate $\hat{\mathbf{d}}$. By construction $\delta = 1$ at $u = 1.5$, and $\delta > 1$ wherever the raw score is smaller, so it is a multiplier on exposure rather than a fraction of it. Table 2 does exactly that, holding the Fisher score equal to seven digits by tuning the exposure factor δ . The loss still spans a factor 9.4.

The two objectives also disagree on which input to prefer: the raw score, before δ equalises it, is largest at $u = 0$, which is why δ is smallest there, and the loss is 1.7 times its value at $u = 1$. How far they disagree is a property of the instance; what is structural is only the blindness.

6.3. The reachability floor with a single input

With $m = 1 < d - 1$ no direction lies on every reachable set at once. Along the closed loop at q^* the floor is largest at the start of life, where $\beta_{\min} = 4.52^\circ$; it falls through mid-life to below the 0.01° resolution of our sampling, where the indicator is momentarily reachable, and rises again towards failure

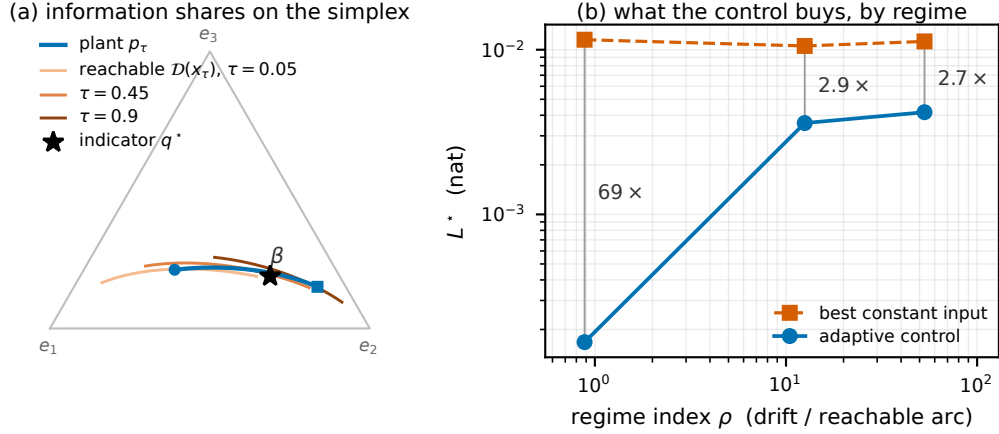


Figure 4: (a) The information-share curve p_τ on the 2-simplex (circle: start, square: failure), the reachable sets $\mathcal{D}(x_\tau)$ at $\tau = 0.05, 0.45$ and 0.9 , and the optimal indicator q^* . No point lies on all three, so β cannot be held at zero (Theorem 4.1). (b) What steering buys, against the regime index ρ .

755 (Fig. 5(a)). Zero average loss would need the indicator reachable at every
 756 instant and not at one, so it is unattainable however the input is steered, in
 757 agreement with Theorem 4.1. Fig. 4(a) shows the geometry: the reachable
 758 set sweeps across q^* as the plant ages, so the mid-life arc passes close to the
 759 indicator while the arcs at $\tau = 0.05$ and $\tau = 0.9$ stand off it.

760 6.4. The counting rule, tested in both directions

761 Theorem 4.1 is an equivalence, and the paragraph above tests only half
 762 of it: with too few inputs the floor stays positive. The other half, that
 763 supplying the inputs the counting rule asks for makes the floor vanish, is the
 764 more useful claim for a practitioner deciding how many operating variables
 765 to instrument, so we test it directly.

766 Give the same three-mechanism plant a second independent input by
 767 writing $\varphi_i(u) = \exp(\theta_i - E_i u_1 - G_i u_2)$ with $G = (0.9, 0.3, 1.2)$, so that $m =$
 768 $2 = d - 1$ and each input acts on the mechanisms in a different pattern.
 769 Then $\log \mathbf{d}_i$ is affine in (u_1, u_2) , and matching a target direction fixes the
 770 $d - 1 = 2$ differences of logarithms: two linear equations in two unknowns,
 771 with a coefficient matrix of determinant 0.27 here, hence a unique solution
 772 at every state.

773 Taking as target the direction the plant occupies at mid-life under a
 774 central operating point, that solution lies inside the admissible box $[0, 3]^2$

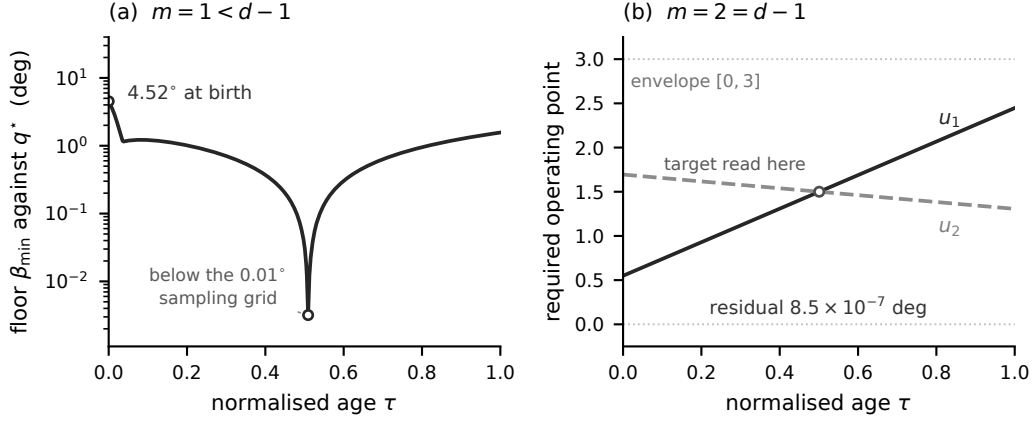


Figure 5: (a) With $m = 1$ the floor against q^* is largest at birth and dips below the sampling grid near mid-life. (b) With $m = 2$ the mid-life direction is held exactly; plotted is the operating point this requires, which never leaves $\mathcal{U}$.

at every one of 600 sampled states covering the whole life, and the residual misalignment there is 8.5×10^{-7} degrees, that is, zero to numerical precision; the schedule it requires is drawn in Fig. 5(b). An independent 90×90 grid search over the box returns a floor of at most 0.156° , which is the resolution of the grid rather than a property of the plant.

The contrast with the single-input case is the content of the theorem: the same mechanisms, the same shapes, the same envelope on each input, and the peak floor moves from 4.52° to zero at every sampled state on the addition of one operating variable. On this plant the counting rule is therefore not a conservative sufficient condition but a sharp one, and within the adopted model it fixes how many independent control degrees of freedom the plant must offer. How those degrees of freedom correspond to physical actuators, and how the mechanisms are individuated in the first place, are modelling questions this paper does not settle. Proposition 4.4 says which plants escape it, and Remark 4.5 says why a designer should not count on being one of them.

6.5. The value of steering

Minimising over q gives $q^* = (0.466, 0.771, 0.435)$, at which the adaptive policy attains 1.68×10^{-4} against 1.15×10^{-2} under the best constant input, a factor of 68. The ratio depends on how finely $\mathcal{U}$ is discretised for the policy, rising to 72 at 161 points; we report it on the 41-point grid used throughout.

796 The factor is a ratio of information losses, not of information, and it is
797 not free in reliability terms: the policy that attains it runs the component
798 differently, and Section 6.6 finds the two closed loops separated by a fac-
799 tor of 3.4 in life. Aligning the indicator and prolonging the component are
800 competing objectives, which is what makes the horizon in (24) a controlled
801 quantity rather than a constant. That factor is the value of steering at all,
802 not the value of steering optimally: a myopic policy that attends only to the
803 numerator of (24) sits 2.8% above the optimum at q^* .

804 The fraction in (24) is doing visible work at this operating point. The
805 closed loop at q^* and the closed loop at a balanced indicator, which calls for
806 far less steering, differ by a factor 3.4 in lifetime: what alignment buys in the
807 numerator is paid for in the denominator. This is the tension anticipated in
808 Section 5, and it is why the horizon cannot be fixed in advance.

809 *6.6. The computed fractional optimum*

810 Dinkelbach bisection with the control grid matched to the policy gives,
811 over $N \in \{25, 29, 35\}$, extrapolants agreeing to 1.2% and $\lambda_\infty^* = 1.63 \times 10^{-4}$,
812 below the realisable 1.68×10^{-4} as it must be.

813 We quote this to two digits only, for the following reason: the raw values
814 are twice the extrapolant, so the first-order correction carries half the answer,
815 and both quantities still move by 4% between control grids of 41 and 81
816 points. Three checks guard the number: F must decrease and cross zero, λ^*
817 must not exceed a realisable policy, and the grid sequence must extrapolate
818 consistently. A fixed-horizon formulation failed the second and third and was
819 discarded, as did a first attempt with mismatched control grids. A converse
820 bound uniform over trajectories, which would replace this computation by a
821 theorem, remains open.

822 *6.7. The regime index, and how far it screens*

823 For a practitioner the first question is not how to steer but whether to
824 bother. One dimensionless ratio answers it. The reachable set $\mathcal{D}(x)$ is an
825 arc; write ρ for the drift of its midpoint over the life divided by its mean
826 length, so ρ measures how far the informative direction wanders in units of
827 the actuation authority available to chase it.

828 Table 3 varies the spread of the activation energies, which by (20) is what
829 sets that authority. As the arc collapses, the attainable loss L^* rises by a
830 factor 25 and the value of control falls from 68 to 2.7: with weak authority
831 there is little left to steer. Fig. 4(b) plots the two against ρ . The construction

Table 3: The regime index. The activation energies E set the actuation authority; c , the noise and $\mathcal{U}$ are held fixed. L^* is the loss under adaptive control, and the gain its ratio to the best constant input.

E	arc	ρ	L^*	gain
(0.60, 1.10, 0.80)	33.9°	0.88	1.7×10^{-4}	68×
(0.80, 0.82, 0.79)	1.93°	12.5	3.6×10^{-3}	2.9×
(0.80, 0.805, 0.798)	0.45°	53.1	4.2×10^{-3}	2.7×

832 of this paper is a statement about the low- ρ regime, and ρ can be computed
833 from a nominal degradation model before any indicator is designed.

834 Three plants differing only in their activation energies are not a screening
835 law, so the index was tested on 299 further plants drawn at random: two to
836 five mechanisms, one or two independently adjustable inputs, and the pre-
837 exponentials, activation energies, shape coefficients, input envelope, initial
838 state and signal-to-noise ratio varied together. The rank correlation between
839 ρ and the gain is -0.36 , with a bootstrap 95% interval from -0.46 to -0.24 ,
840 and it keeps its sign inside every stratum of the mechanism count and of the
841 input count, so it is not an artefact of either.

842 Read as a classifier of whether steering repays a factor of five, which 32%
843 of the plants do, ρ has an area under the receiver-operating curve of 0.73,
844 with interval 0.67 to 0.79. The reachable arc alone scores 0.73 as well, and
845 the drift alone 0.53, with interval 0.46 to 0.60, which includes the 0.5 of a
846 coin toss: most of the index’s discriminating power is in its denominator, and
847 the drift earns its place only among plants of equal arc. Of the plants below
848 $\rho = 2.6$, 52% repay steering fivefold, against 14% above it, a split read off
849 this same sample and so optimistic. As a veto the index is weaker than that
850 suggests: rejecting every plant with $\rho \geq 1$ would also reject 23% of plants
851 that would in fact have repaid a factor of five, and that is the expensive error
852 rather than the cheap one. Nor is lower always better. In the lowest tenth of
853 ρ only 43% repay fivefold, because there the drift itself is small, a median of
854 2.7° against 14° over the sample: steering needs authority, and it also needs
855 something to chase. The first row of Table 3 is not typical of its own regime
856 either: among sampled plants within a factor 1.4 of its ρ , the median gain is
857 5.2 rather than 68. ρ describes a regime; it does not decide a design.

858 *6.8. What the loss buys a downstream estimator*

859 Corollary 2.4 prices the loss in the error of the estimate the model itself
 860 prescribes. A practitioner does not use that estimate, so the ordering is worth
 861 checking against a procedure that uses none of the construction: sample the
 862 indicator over the last 40% of life, match the observed history against the
 863 indicator’s own reference curve, and read remaining life off the best-fitting
 864 time offset. That estimator sees a scalar signal and nothing else — not $\hat{\mathbf{d}}$,
 865 not γ , not the loss.

866 Over twelve indicators spanning $\bar{\ell}$ from 0.012 to 0.37, its median absolute
 867 error — one of the standard prognostic accuracy measures [27] — rises with
 868 the loss, at Spearman correlation 0.89, and the best-aligned indicator beats
 869 the worst by a factor 1.5. Twelve indicators is a small sample, so both
 870 are given with intervals obtained by resampling the 160 units behind each
 871 median: 0.34 to 0.90 for the correlation, and 1.40 to 2.03 for the factor. The
 872 ordering is not in doubt — a one-sided permutation test gives $p = 1.5 \times 10^{-4}$
 873 — but its strength is, and the point estimate sits near the top of its own
 874 interval. The factor is far smaller than the 31-fold spread in $\bar{\ell}$, and it should
 875 be: the loss is a logarithmic information quantity, the error is a time, and
 876 an estimator with the whole history recovers much of what any one instant
 877 discards.

878 The same protocol on 120 further plants, drawn at random as in Sec-
 879 tion 6.7, gives correlations between 0.87 and 0.99, with the plant above
 880 among the weaker. That design tilts every indicator toward the same di-
 881 rection, however, so loss and error both grow with one parameter and their
 882 agreement is close to structural. Tilting instead by a random amount of
 883 up to 60° in a random direction weakens the median correlation to 0.74,
 884 with the interval excluding zero on 84% of plants. It also exposes what the
 885 experiment cannot show. On these plants the angle between an indicator and
 886 the mid-life informative direction ranks the indicators identically to the loss
 887 on 82% of plants, with neither consistently better on the rest, because the
 888 direction rotates little beside how far the indicators differ. The experiment
 889 is therefore evidence that the ordering of the loss carries over to an estimator
 890 built from none of this construction; it is not evidence that the time-averaged
 891 loss improves on a mid-life snapshot of the misalignment, which would need
 892 a plant whose direction rotates by more than its candidate indicators differ.

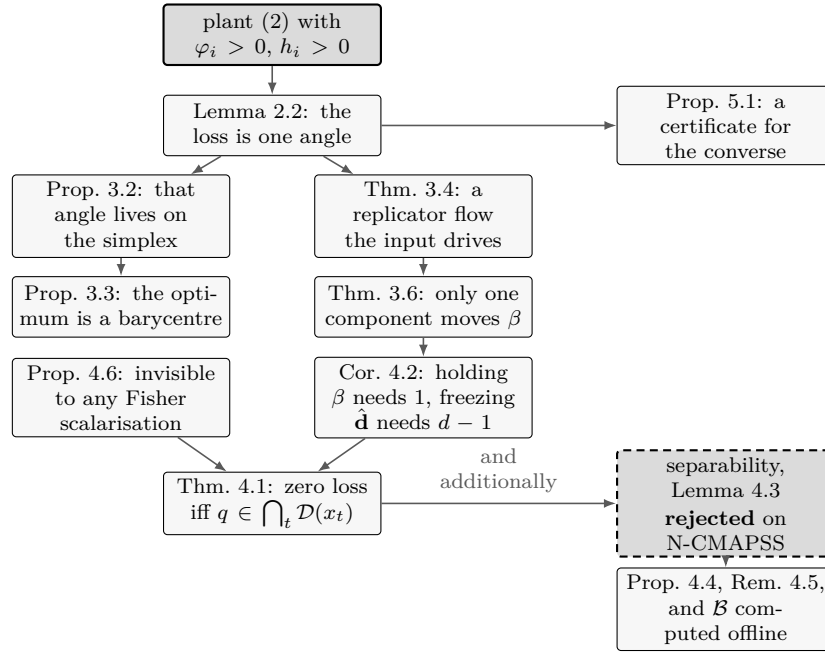


Figure 6: What rests on what. The spine needs only the positivity of the degradation model. Only the dashed branch adds separability, which Section 7 reports as rejected on the best public turbofan data.

893 7. What public data can say about the premises

894 The construction rests on three modelling choices, and each is empirically
 895 testable. They do not carry equal weight, and Fig. 6 separates what depends
 896 on which. Two are premises about the plant: that the informative direction
 897 rotates along the trajectory, and that the input moves it. The third is the
 898 separability of Lemma 4.3, which is what makes the operator’s authority
 899 computable offline. Table 4 summarises what six public run-to-failure fleets
 900 currently settle. Supplementary Section S1 gives the protocol behind every
 901 row, with the parameters of the code that produced it.

902 Three readings qualify the table. The Severson channels are capacity
 903 fade, resistance growth and mean temperature, and mean temperature is a
 904 monitoring channel rather than a damage state — it is the operating vari-
 905 able itself in the Arrhenius form of Section 2. That row therefore reads the
 906 rotation of a direction formed from three measured series, which is what the
 907 premise asserts, and does not need (5)’s pairing of channels with mecha-
 908 nisms. On the four measured fleets the obstacle to testing actuation is the

Table 4: What the six public fleets settle, and what they cannot. Only the two turbofan simulations vary the operating point within a unit, which is why the actuation premise is confirmed in simulation and open on measured hardware.

fleet	premise	what is measured	reading
Severson lithium- ion [28]	rotation	arc of an empirically constructed informative direction, the fleet mean, over $\tau \in [0.15, 0.9]$, on 129 Severson lithium-ion cells	travels 30.7° of arc and ends 21° from its start (Fig. 2(c)); stays in the open positive orthant once the window reaches 301 cycles
XJTU- SY [29], PRONOS- TIA [30]	actuation	—	the operating condition is fixed per unit, so condition and group of units cannot be separated
NASA batteries [31]	actuation	—	four cells
C- MAPSS [32]	actuation	between-regime minus within-regime angle, each engine serving as its own control	by 7.9° (FD004) and 10.8° (FD002), positive for 436 of 439 engines; fake regimes give $ B - W \leq 0.5^\circ$
N-CMAPSS DS02, DS01 [33]	separability	interaction against both main effects, in a two-way layout inside engines	larger by a factor 1.65 over the operating-point effect on DS02 and 1.84 on DS01, under every clustering and staging tried: rejected

estimator and not the physics, so the premise is confirmed in simulation and open on hardware. Both the Severson and the C-MAPSS rows estimate the direction by differentiating each channel along normalised life, which reads a time derivative as the sensitivity of Eq. (6) and so presumes the time-shift population of Section 2. And the C-MAPSS contrast is reported against a negative control — fake regimes cut from a single real one, where no contrast exists by construction — because a procedure that returned a positive contrast regardless would produce the same table entry.

The consequence is the scope of the results above. The geometry of Sections 3 and 4 needs only $\varphi_i > 0$ and $h_i > 0$, and is untouched. What separability buys, and what the rejection withdraws, is Proposition 4.4 and the offline route to $\mathcal{B}$: those describe a model class, and a plant must be shown to belong to it before they are used. The two-way layout above is the test, and it costs one pass over data a reliability programme already holds.

923 8. Discussion

924 Four practical readings follow from the results above. The first two hold
 925 whatever the plant is, because one is a test and the other an invariance. The
 926 last two assume the separable form of (2), which Section 7 reports as rejected
 927 on the only data able to test it, so they apply where that form does and not
 928 otherwise.

929 Test the separability assumption before relying on it. The two-way layout
 930 of Section 7 needs only a fleet whose operating point varies within a unit,
 931 and costs one pass over data a reliability programme already holds. If the
 932 interaction clears the measured noise floor, the offline route to the operator’s
 933 authority is unavailable and the two model-based readings below do not
 934 apply.

935 No Fisher criterion can rank fixed indicators by the loss they incur. This
 936 is the second of the two unconditional readings: Proposition 4.6 is an in-
 937 variance of the loss under a common gain and holds whatever the plant, so
 938 rules that maximise a scalarisation of the Fisher information cannot separate
 939 configurations whose losses differ by an order of magnitude. Where the two
 940 objectives do coincide, that is a property of the application, to be verified
 941 rather than assumed.

942 The binding constraint is actuation authority, not indicator sophistica-
 943 tion. Theorem 4.1 prices the operating variables a lossless fixed indicator
 944 would need, and Proposition 4.4 sharpens the count to sufficient but not
 945 necessary, with exceptions that Remark 4.5 shows to be a knife edge. Care
 946 in choosing the weights substitutes for actuation in neither case. What one
 947 controllable variable does buy is the weaker objective of Corollary 4.2: an
 948 installation that could never freeze the informative direction can still hold
 949 the misalignment constant, though not at zero. Authority bought this way
 950 is not free, and Section 6.6 prices it in life.

951 The decision to steer precedes the design of the indicator. The regime
 952 index ρ of Section 6.7 requires only a nominal degradation model and the
 953 operating envelope, and Table 3 shows the value of steering collapsing from
 954 $68\times$ to under $3\times$ as the reachable arc shortens relative to the drift. It
 955 indicates a regime rather than deciding a design: Section 6.7 measures how
 956 often it is wrong, and in which direction.

957 *Limitations.* Four limitations qualify the readings above. The separable form
 958 of (2) is not only an idealisation but one the data reject: Section 7 finds

959 the interaction between operating point and state larger than either main
 960 effect on two N-CMAPSS datasets, so computing the operator's authority
 961 offline is available only where that form holds. The optimum of Section 5
 962 is computed rather than certified, to two digits. The actuation premise is
 963 confirmed on simulated rather than measured data, for the reasons set out in
 964 Section 7. And the population is taken to slide along one degradation path,
 965 which is what makes remaining life a coordinate on it; under dispersion in
 966 the coefficients of (2) the sensitivity that Remark 2.1 measures stands some
 967 45° from (6), so the construction describes fleets that vary in when they fail
 968 rather than in how they degrade.

969 **9. Conclusion**

970 A fixed health indicator discards information, and this paper has treated
 971 what it discards as a quantity the operating point controls. Indicator design
 972 then becomes a reachability question. The family of reachable direction sets
 973 bounds what the operating point can achieve, and their drift relative to their
 974 width is a dimensionless measure, available before any indicator is fixed,
 975 of whether substantial benefit is plausible. Actuation authority, rather than
 976 sophistication of the indicator, governs whether a fixed compression can come
 977 close to lossless. Whether that authority is present is an empirical question
 978 rather than a modelling assumption, and on 439 simulated turbofan records
 979 it is. Establishing the same on measured fleets awaits records in which the
 980 operating point varies enough to show the effect.

981 Section 7 rejects that separable form on the highest-fidelity public tur-
 982 bofan data, so the geometry of Section 4.1, and the offline route it opens,
 983 describe a model class rather than measured machinery. What replaces that
 984 structure when the stress model is not separable, and whether a weaker one
 985 still yields a computable reachability criterion, remains open.

986 Two further questions remain open, the first of them now in a more
 987 specific form. Proposition 5.1 reduces the search for a converse bound uni-
 988 form over trajectories to the exhibition of a single continuously differentiable
 989 verification function; the value function is the natural candidate and fails
 990 the differential test for a structural reason, so what is required is a smooth
 991 under-approximation of it, by sum-of-squares programming or by interval
 992 arithmetic with an explicit Lipschitz constant. The other is unchanged: the
 993 computed optimum to better than two digits, which requires a scheme of
 994 higher order than the semi-Lagrangian one used here.

995 **Data availability and reproducibility**

996 The six fleets named in Section 7 are publicly available. One script
997 recomputes every reported number from a single configuration and com-
998 pares it with the value written in this manuscript and in its supplemen-
999 tary material, covering the tables and the quantities reported alongside
1000 them; all of its checks pass, and the output of the run that verified this
1001 version is included. The regime-index sweep of Section 6.7 takes hours,
1002 so it and the downstream repeat of Section 6.8 are recorded by two fur-
1003 ther scripts, and the checks read the records. Three groups of numbers lie
1004 outside the checks: the residuals of the identity checks in Supplementary
1005 Section S4, the barycentre comparison in Section 3.3, and the diagnostics
1006 of the two candidate certificates in Supplementary Section S3. The plots
1007 are produced by three further scripts. All of these are available at [https:](https://github.com/HuyHoangLe0201/health-indicator-information-loss)
1008 [//github.com/HuyHoangLe0201/health-indicator-information-loss](https://github.com/HuyHoangLe0201/health-indicator-information-loss).

1009 **Declaration of competing interest**

1010 The authors declare that they have no known competing financial inter-
1011 ests or personal relationships that could have appeared to influence the work
1012 reported in this paper.

1013 **Funding**

1014 This research did not receive any specific grant from funding agencies in
1015 the public, commercial, or not-for-profit sectors.

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